

KAZYR, Russian Federation

WMO: 298760

Lat: 53.6861N

Lon: 94.2042E

Elev: 1464

StdP: 13.94

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.4	-17.7	-29.2	1.1	-22.9	-22.7	1.7	-17.4	15.0	4.6	11.0	9.1	0.1	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	86.1	66.1	82.4	64.5	78.7	63.0	68.8	80.5	66.8	77.2	65.1	74.7	2.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	98.4	72.9	63.5	92.5	70.7	61.8	87.2	68.6	34.0	80.9	32.4	77.2	31.0	74.4	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	8.0	6.9	-29.9	93.1	8.4	4.2	-35.9	96.2	-40.8	98.6	-45.5	101.0	-51.6	104.0		
(5)			-31.4	72.1	7.7	2.6	-36.9	74.0	-41.4	75.6	-45.7	77.0	-51.3	78.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	7.5	12.5	26.4	38.4	48.0	60.7	65.3	62.3	50.3	36.3	19.1	11.9
	DBStd	22.06	11.31	11.18	10.51	7.96	8.31	8.07	5.24	6.16	7.90	8.16	11.73	11.02
	HDD50	6234	1316	1049	731	355	143	10	0	1	92	430	926	1181
	HDD65	10491	1781	1469	1196	798	530	174	61	126	444	890	1376	1646
	CDD50	1380	0	0	0	7	80	332	473	383	100	5	0	0
	CDD65	162	0	0	0	0	2	46	69	43	2	0	0	0
	CDH74	2019	0	0	0	6	106	766	686	401	54	0	0	0
	CDH80	595	0	0	0	0	19	258	196	112	10	0	0	0
	WSAvg	1.9	2.0	1.8	2.1	2.2	2.1	1.7	1.5	1.3	1.6	1.7	2.2	2.1
	PrecAvg	29.9	1.1	1.0	1.3	2.0	2.6	3.4	4.2	4.3	3.0	2.6	2.6	1.9
(15) Precipitation	PrecMax	36.2	1.9	3.0	3.0	3.3	4.6	5.0	7.1	7.4	5.5	7.6	4.4	3.7
	PrecMin	23.7	0.4	0.3	0.4	0.8	0.7	1.7	2.0	2.5	1.0	0.8	1.5	0.5
	PrecStd	3.1	0.4	0.5	0.7	0.7	0.9	0.9	1.0	1.0	1.1	1.2	0.7	0.7
	PrecStd	3.1	0.4	0.5	0.7	0.7	0.9	0.9	1.0	1.0	1.1	1.2	0.7	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.5	41.0	58.3	70.9	82.4	90.9	89.1	88.6	80.6	65.8	45.7	34.1
		MCWB	27.5	32.1	43.0	51.4	58.4	67.9	67.9	68.8	60.6	51.0	40.2	31.3
	2%	DB	27.4	34.4	51.2	63.8	76.8	86.2	85.6	82.7	72.9	58.3	40.3	31.2
		MCWB	24.8	29.4	39.1	48.1	57.1	65.3	66.5	66.7	58.9	47.7	36.1	28.4
	5%	DB	25.3	30.9	45.5	58.4	71.0	82.5	81.5	78.2	67.4	51.9	35.2	28.3
		MCWB	22.7	27.6	36.0	45.3	53.1	64.0	65.1	64.4	56.7	44.0	31.9	25.9
	10%	DB	22.5	27.7	40.9	52.9	65.0	77.9	77.5	74.0	62.0	47.1	32.5	26.0
		MCWB	19.8	24.5	33.7	42.0	51.5	61.7	64.2	62.8	53.9	42.1	30.2	24.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.6	33.4	43.9	53.5	61.7	72.4	71.0	70.7	62.8	52.7	41.2	32.9
		MCDB	29.6	37.7	55.5	65.3	78.9	87.0	83.1	82.6	74.2	63.4	44.4	34.3
	2%	WB	25.5	30.6	39.9	49.3	58.2	68.3	68.4	68.3	60.0	48.7	36.4	28.6
		MCDB	27.0	33.5	49.5	61.8	71.8	79.9	80.4	78.7	70.5	56.7	39.4	30.1
	5%	WB	22.7	27.8	36.8	45.9	55.5	65.5	66.9	66.1	57.7	45.7	32.7	26.3
		MCDB	25.2	30.9	44.5	55.9	67.6	77.4	77.1	75.6	66.3	50.8	34.8	28.0
	10%	WB	19.8	24.2	34.1	43.3	52.6	63.5	65.3	64.1	55.1	42.7	30.4	24.2
		MCDB	22.2	27.6	40.1	52.1	62.5	75.3	74.2	71.5	60.9	46.5	32.4	25.9
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	20.6	22.7	21.2	22.6	24.6	20.6	20.2	19.3	14.6	11.9	13.1
		MCDBR	13.2	21.0	27.5	31.1	35.3	34.0	29.6	29.0	29.3	25.0	13.0	11.4
		MCWBR	11.3	15.4	16.9	17.1	17.3	14.4	12.3	13.3	15.0	14.9	9.4	10.3
	5% WB	MCDBR	12.2	18.2	25.1	29.0	31.0	29.1	24.7	25.7	27.3	22.9	10.9	10.5
		MCWBR	10.9	13.3	15.6	16.6	17.0	13.6	11.8	12.9	14.3	14.3	8.6	9.9
(40) Clear-Sky Solar Irradiance	taub		0.296	0.336	0.383	0.426	0.433	0.409	0.418	0.407	0.373	0.359	0.336	0.305
	taud		1.985	1.929	1.864	1.933	2.101	2.314	2.318	2.316	2.339	2.194	1.977	1.925
	Ebn at Noon		195	226	242	251	258	267	262	258	252	221	173	158
	Edh at Noon		25	37	49	52	47	38	38	36	31	28	25	22
(44) All-Sky Solar Radiation	RadAvg		327	648	1067	1388	1590	1820	1660	1381	951	540	285	214
	RadStd		32	52	83	75	158	201	127	120	122	53	42	24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page